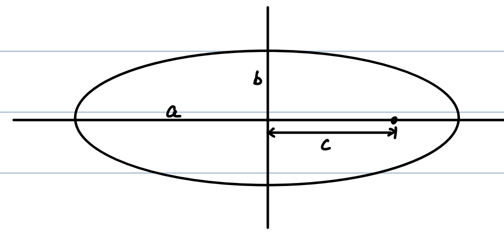
Law of Gravity

Kepler's laws Johannes Kepler (1571–1630)

1. The law of orbits:

All planets move in elliptical orbits, with the Sun at one focus.



a (semimajor, 半长轴)

b (semiminor, 半短轴)

c (linear eccentricity, 半焦距)

2. The law of areas:

Radial vector connecting Sun and a planet sweeps out equal areas in equal time intervals.

$$\frac{dA}{dt} = C$$

3. The law of periods:

The square of the period $\propto$ cube of semimajor axis.

$$T^2 \propto a^3$$

Only one step away from the truth:

1 & 2 $\Rightarrow$ central force (conservation of angular momentum)

3 $\Rightarrow$ magnitude of the force

Suppose $a = b = r$ (circle)

$$\left\{ \begin{array}{l} T^2 \propto r^3 \\ v = \frac{2\pi r}{T} \end{array} \right. \Rightarrow v \propto \frac{1}{\sqrt{r}}$$

$$\Rightarrow a = \frac{v^2}{r} \propto \frac{1}{r^2}$$

$$\Rightarrow F = ma \propto \frac{m}{r^2}$$

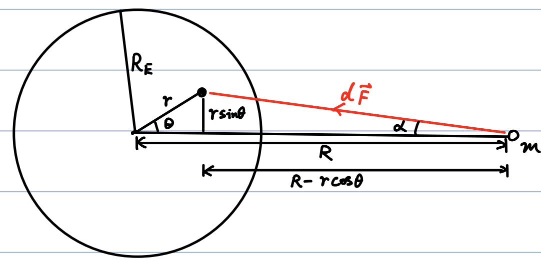
- Newton's law of Universal Gravitation:

$$\boxed{F = G\frac{m_1 m_2}{r^2}} \qquad G \approx 6.67 \times 10^{-11}\ N\cdot m^2/kg^2$$

Principle of superposition for gravity

$$\vec{F}_{1,net} = \vec{F}_{12} + \vec{F}_{13} + \cdots + \vec{F}_{1n}$$

Earth as a finite-size ball :



① $R > R_E$

for small volume : $dV = r^2 \sin\theta \, dr \, d\theta \, d\varphi$

$$dF = \frac{G \rho dV \cdot m}{r^2 \sin^2\theta + (R - r\cos\theta)^2}$$

$$\Rightarrow F = \int dF \cos\alpha$$

$$= \int \frac{G \rho r^2 \sin\theta \, dr d\theta d\varphi \cdot m}{r^2 \sin^2\theta + (R - r\cos\theta)^2} \cdot \frac{R - r\cos\theta}{[r^2\sin^2\theta + (R - r\cos\theta)^2]^{1/2}}$$

$$= G\rho m \cdot 2\pi \int_0^{R_E} dr \int_0^{\pi} d\theta \, \frac{r^2 \sin\theta \, (R - r\cos\theta)}{[r^2 \sin^2\theta + (R - r\cos\theta)^2]^{3/2}}$$

$$= 2\pi G\rho m \int_0^{R_E} dr \int_{-1}^{1} d(\cos\theta) \, \frac{r^2 (R - r\cos\theta)}{[R^2 + r^2 - 2Rr\cos\theta]^{3/2}}$$

$$= 2\pi G \rho m \int_0^{R_E} dr \cdot \frac{2r^2}{R^2}$$

$$= 2\pi G\rho m \cdot \frac{2}{R^2} \cdot \frac{1}{3} R_E^3$$

$$= \frac{GMm}{R^2} \qquad M \equiv \rho \cdot \frac{4}{3}\pi R_E^3$$

$\Rightarrow$ Earth can be treated as point particle.

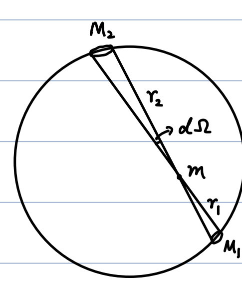
$$F = mg \Rightarrow g = \frac{GM}{R_E^2}$$

② $R < R_E$

Consider an empty shell

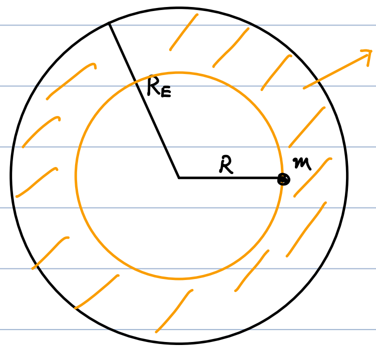
$M_1 = \lambda r_1^2 d\Omega$

$M_2 = \lambda r_2^2 d\Omega$



$$\Rightarrow \begin{cases} F_1 = \dfrac{GM_1 m}{r_1^2} = \lambda m\, d\Omega \\ F_2 = \dfrac{GM_2 m}{r_2^2} = \lambda m\, d\Omega \end{cases}$$

cancel!

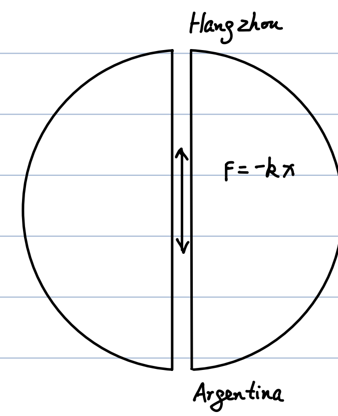


No contribution!

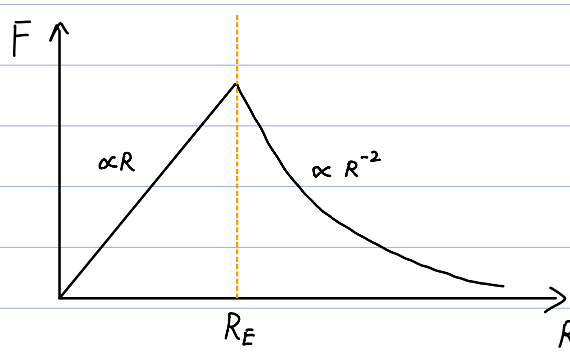
$$F = \frac{GM'm}{R^2}$$

$$M' = \rho \cdot \frac{4}{3}\pi R^3 = \frac{M}{\frac{4}{3}\pi R_E^3} \cdot \frac{4}{3}\pi R^3 = \left(\frac{R}{R_E}\right)^3 M$$

$$\Rightarrow F = \frac{Gm}{R^2} \cdot \frac{R^3}{R_E^3} M = \frac{GMm}{R_E^3} R \quad (\text{spring})$$



In total:



- Prediction of "universal gravitation":

~~Sun~~ Earth $\longleftrightarrow$ ~~Earth~~ Moon

$$\begin{cases} F = \dfrac{GM_E M_M}{r_M^2} = M_M \dfrac{v_M^2}{r_M} \\ \dfrac{GM_E}{R_E^2} = g \end{cases}$$

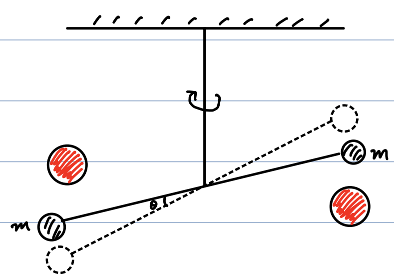
$$\Rightarrow \left.\begin{aligned} \frac{gR_E^2}{r_M} &= v_M^2 \\ v_M &= \frac{2\pi r_M}{T} \end{aligned}\right\} \Rightarrow r_M^3 = \frac{gR_E^2 T^2}{4\pi^2}$$

$$\begin{cases} g \approx 9.8\ m/s^2 \\ R_E \approx 6.4 \times 10^6\ m \\ T \approx 27.3215\ \text{days} \approx 2.361 \times 10^6\ s \\ r_M \approx 3.84 \times 10^8\ m \end{cases}$$

LHS : $r_M^3 \approx 5.66 \times 10^{25}\ m^3$

RHS : $\frac{g R_E^2 T^2}{4\pi^2} \approx 5.67 \times 10^{25}\ m^3$ ✓

- Measurement of G Henry Cavendish (1731 ~ 1810)



$\Rightarrow 6.754 \times 10^{-11}\ m^3/(kg \cdot s^2)$

(@ 1798)

- Density of Earth :

$$\frac{G M_E}{R_E^2} = g$$

$$\Rightarrow \begin{cases} M_E = \frac{g R_E^2}{G} \approx \frac{9.8 \times (6.4 \times 10^6)^2}{6.67 \times 10^{-11}} \approx 6.02 \times 10^{24}\ kg \\ V_E = \frac{4}{3}\pi R_E^3 \approx 1.10 \times 10^{21}\ m^3 \end{cases}$$

$$\Rightarrow \rho_E = \frac{M_E}{V_E} \approx 5.5 \times 10^3\ kg/m^3$$

reference : coal : 1.1 ~ 1.4

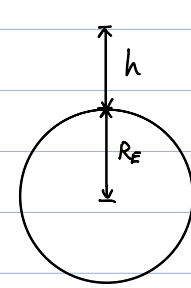
marble : 2.4 ~ 2.7

iron : 7.9

$\Rightarrow$ high density near center of earth

- Modification of gravity near surface of Earth :

$$F = \frac{G M_E m}{(R_E + h)^2}$$



if $h \ll R_E$:

$$F = \frac{GMm}{R_E^2} \frac{1}{(1+ h/R_E)^2}$$

$$\approx \frac{GMm}{R_E^2} \left(1- 2\frac{h}{R_E} \right)$$

$$\Rightarrow g \approx \frac{GM}{R_E^2} \left(1- \frac{2h}{R_E} \right) = g_0 \left(1- \frac{2h}{R_E} \right)$$

- Gravitational potential energy

$$U(r) = - \int_{r_0}^{r} F(r')\,dr'$$

$$= \int_{r_0}^{r} \frac{GMm}{(r')^2}\,dr'$$

$$= - \frac{GMm}{r'} \Big|_{r_0}^{r}$$

$$= - \frac{GMm}{r} + C$$

For convenience, set $C=0$

$$\Rightarrow \boxed{U(r) = - \frac{GMm}{r}}$$

check:

$$- \frac{dU}{dr} = - \frac{GMm}{r^2} \equiv F(r) \quad \checkmark$$

- Total mechanical energy :

$$E = K + U = \frac{1}{2} m v^2 - \frac{GMm}{r}$$

- Satellite at r :

$$\frac{GM_E m}{r^2} = m \frac{v^2}{r}$$

$$\Rightarrow v^2 = \frac{GM_E}{r}$$

$$\Rightarrow \boxed{E = - \frac{GM_E m}{2r}}$$

- Geosynchronous orbital :

$$T = 1\,\text{day} = 86400\,s$$

$$\begin{cases} v = \dfrac{2\pi r}{T} \\ v^2 = \dfrac{GM_E}{r} \end{cases}$$

$$\Rightarrow r = \sqrt[3]{\frac{GM_E T^2}{4\pi^2}}$$

$$\approx 4.23 \times 10^7\ m$$

- First cosmic velocity (circular velocity)

$$\frac{1}{2}mv^2 - \frac{GM_E m}{R_E} = -\frac{GM_E m}{2R_E}$$

$$\Rightarrow \quad v = \sqrt{\frac{GM_E}{R_E}} \approx 7.9 \text{ km/s}$$

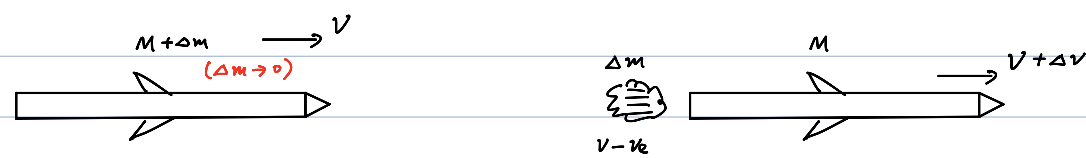
- Second cosmic velocity (escape velocity of Earth)

$$\frac{1}{2}mv^2 - \frac{GM_E m}{R_E} = 0$$

$$\Rightarrow \quad v = \sqrt{\frac{2GM_E}{R_E}} \approx 11.2 \text{ km/s}$$

- Third cosmic velocity (escape velocity of Sun)

- Rocket Propulsion



Momentum conservation (assuming no extra impulse) :

$$(\cancel{M}+\cancel{\Delta m})\, v = M(\cancel{v}+\Delta v) + \Delta m\,(\cancel{v}-v_e)$$

$$\Rightarrow \quad \Delta v = \frac{\Delta m}{M} v_e$$

$$\Rightarrow \quad \boxed{dv = -\frac{dM}{M} v_e}$$

$$\Rightarrow \quad \int_{v_i}^{v_f} dv = -v_e \int_{M_i}^{M_f} \frac{dM}{M}$$

$$\Rightarrow \quad v_f - v_i = +v_e \ln \frac{M_i}{M_f}$$

Notes:

1. larger $\frac{M_i}{M_f}$ → larger $v_f - v_i$
2. Reverse process of perfect inelastic collision. (E_{mec} ↑)
3. We have omitted the impulse from gravity

With gravity.

$$-(M+\Delta m)g\,\Delta t = \left[\, M(\cancel{v}+\Delta v) + \Delta m(\cancel{v}-v_e)\right] - \left[(\cancel{M}+\cancel{\Delta m})\, v\right]$$

$$\Rightarrow \quad -g\,dt = dv + \frac{dM}{M} v_e$$

$$\Rightarrow \quad -g\int_0^{t_f} dt = \int_0^{v_f} dv + v_e \int_{M_i}^{M_f} \frac{dM}{M}$$

$$\Rightarrow \quad v_f = v_e \ln \frac{M_i}{M_f} - g t_f$$

Thrust of rocket (force from the ejected exhausted gas)

$$F_{\text{thrust}} = M\frac{dv}{dt} = -\frac{dM}{dt} v_e$$

$$= v_e \left| \frac{dM}{dt} \right|$$